



THE CIA TRIAD

What is the CIA Triad?

- ▶ The CIA triad is the key principle used to guide the cybersecurity policies used in businesses.
- ▶ The CIA triad stands for Confidentiality, Integrity and Availability.
- ▶ The three ideas were first brought together in 1988
- ▶ It is used to assess the impact of breaches, as well as the effectiveness of system security.

Confidentiality

- ▶ Confidentiality is the secrecy/ privacy of a company's data.
- ▶ This is done through access control of information, and proper authorisation measures.
- ▶ Access control of information is where a company has measures in place that limit who can access the company's data, through different levels of authorisation. This is critical to ensuring that the only people able to access sensitive data are the people who need to.
- ▶ Proper authorisation measures are used to ensure the effectiveness of a company's access control, and can be implemented through access tiers, user credentials and physical keys/ passcards.
- ▶ Confidentiality is critical for the continued success of a business, from maintaining proprietary information to keeping customer's safe from attackers

Confidentiality cont.

- ▶ Confidentiality can be compromised by a series of direct attacks like:
 - ▶ Man-in-the-Middle (MitM) attacks: MitM attacks are where an attacker will insert themselves in a data stream to either steal or alter information before choosing what they want to pass onto the intended source.
 - ▶ Hardware used for logins is stolen by an attacker and used to access confidential data.
- ▶ It can also be compromised without an attack:
 - ▶ Users sharing passwords with friends.
 - ▶ Someone might see another user entering their password (shoulder surfing).

Integrity

- ▶ Integrity is making sure that data is free from tampering.
- ▶ This is done by maintaining the accuracy, authenticity and reliability of data stored by and produced by a company.
- ▶ Common practices to ensure the integrity of data are:
 - ▶ Hashing
 - ▶ Encryption
 - ▶ Digital certificates/ trustworthy Certificate Authorities
 - ▶ Digital signatures
- ▶ Integrity can be verified by non-repudiation, which is where something cannot be denied, such as emails sent with digital signatures

Integrity cont.

- ▶ The integrity of data can be compromised through:
 - ▶ An attacker changing file configurations to allow unauthorised access.
 - ▶ An attacker altering the system logs to hide an attack.
- ▶ It can also be done accidentally:
 - ▶ Someone may make a mistake when entering data.

Availability

- ▶ Availability is the ability for workers or customers to access the data they are allowed to.
- ▶ This is done by ensuring systems, networks and applications all function in the correct way.
- ▶ Availability can be ensured through:
 - ▶ Redundant servers, networks and applications
 - ▶ Applying updates to systems when they become available
 - ▶ Backups and disaster recovery plans

Availability cont.

- ▶ Availability can be compromised from:
 - ▶ Power outages.
 - ▶ Floods/ snowstorms preventing workers accessing important data.
 - ▶ Denial of service attacks (DoS). DoS attacks are when servers are bombarded beyond their capability to handle with fake requests that prevent real requests being processed.
 - ▶ Ransomware. Ransomware is when malware injected into a system encrypts the data on that system and won't decrypt it until the owner of the system pays the attacker a sum of money, often in untraceable ways like Bitcoin.

References

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